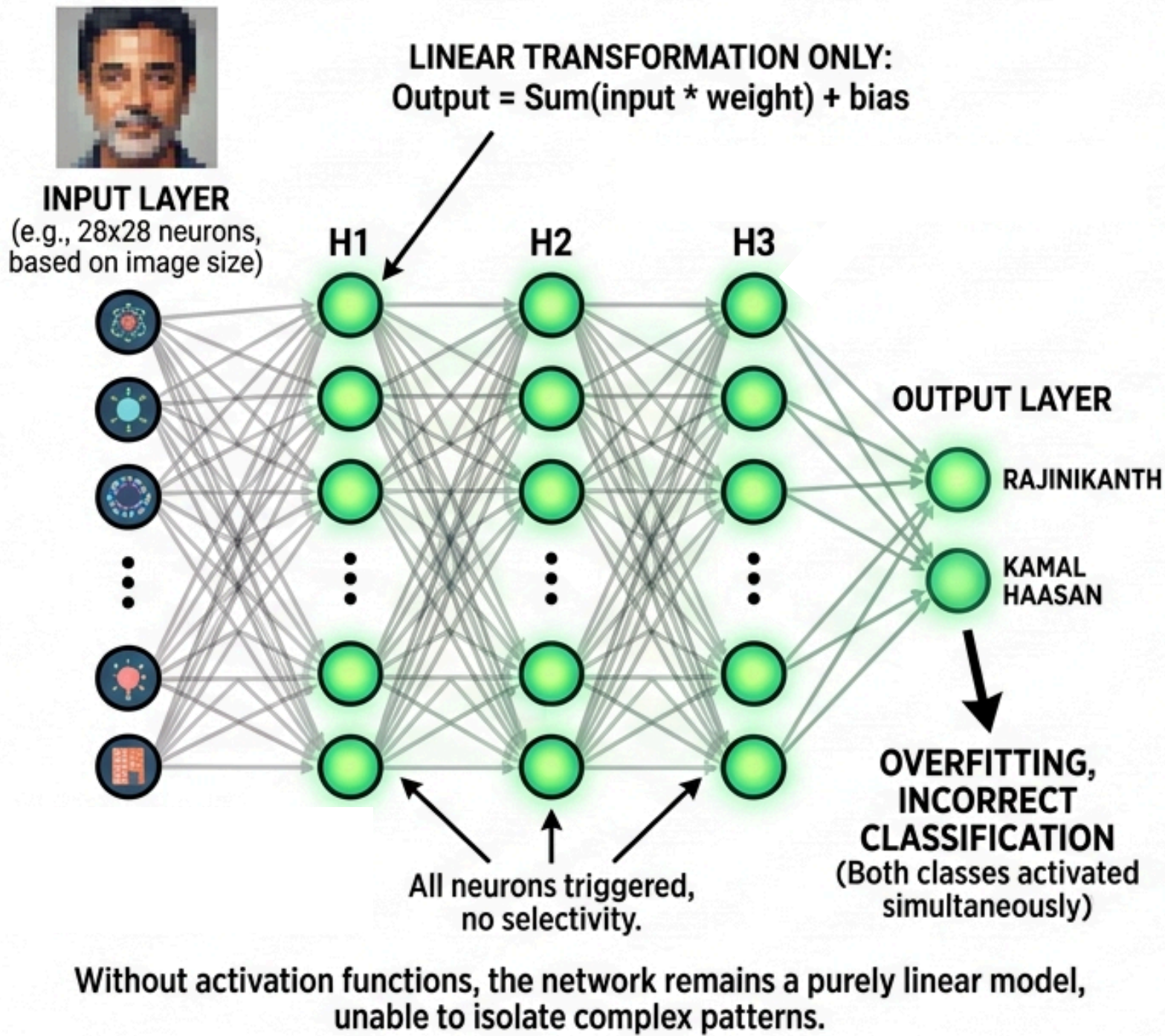


A large, red, circular official stamp with a scalloped edge is centered on the page. The word "OFFICIAL" is written in bold, white, uppercase letters across the center of the stamp. The background is a light gray grid pattern.

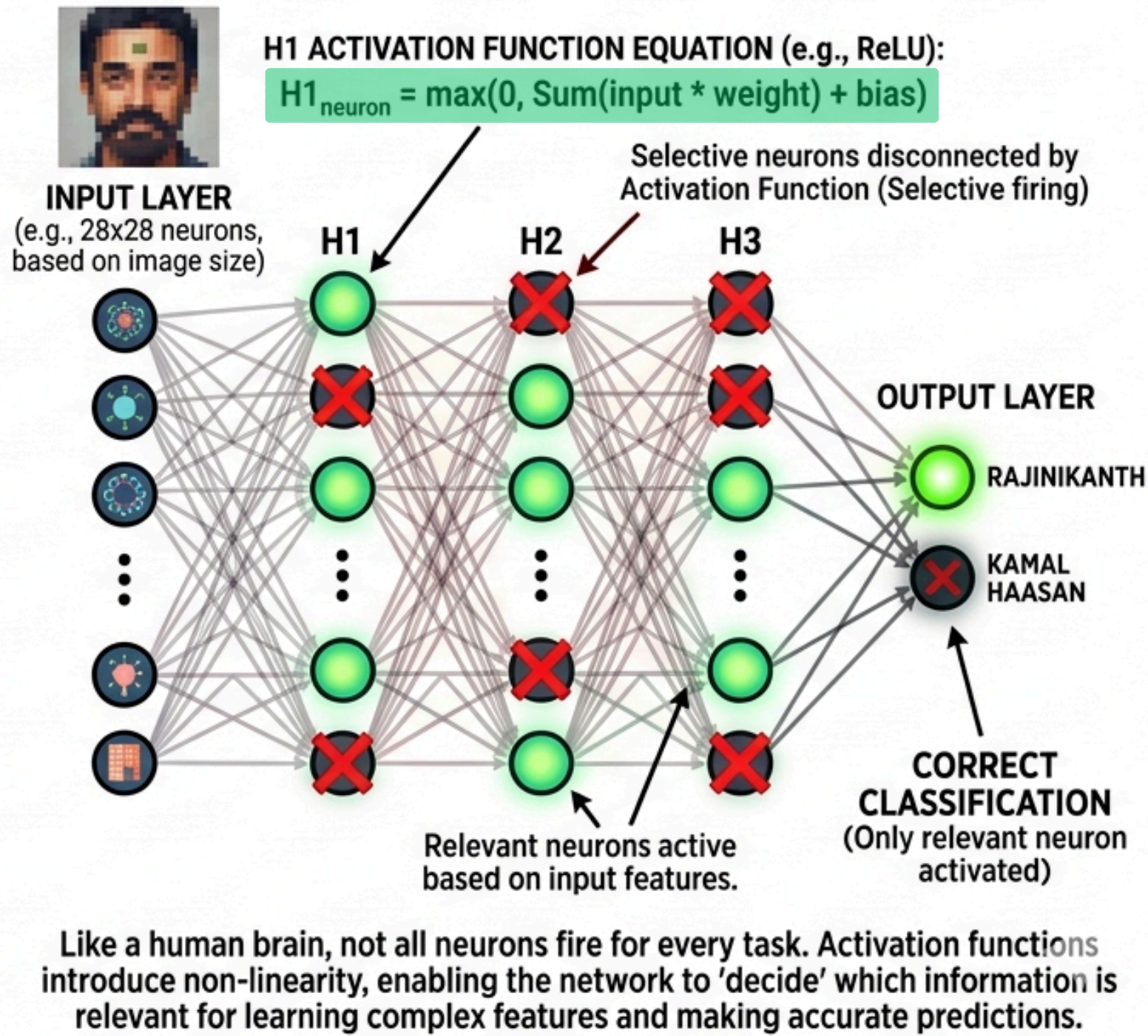
Ready - + 1129

Why Activation Functions?

Neural Network **Without Activation Function**
(All neurons fire, so limited learning)



Neural Network **With Activation Function**
(Selective neurons fire, so EFFECTIVE learning)



Types of Activation Functions

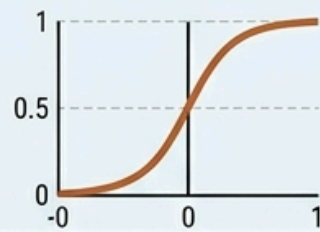
7 COMMONLY Used ACTIVATION Functions

(Non-Linear)

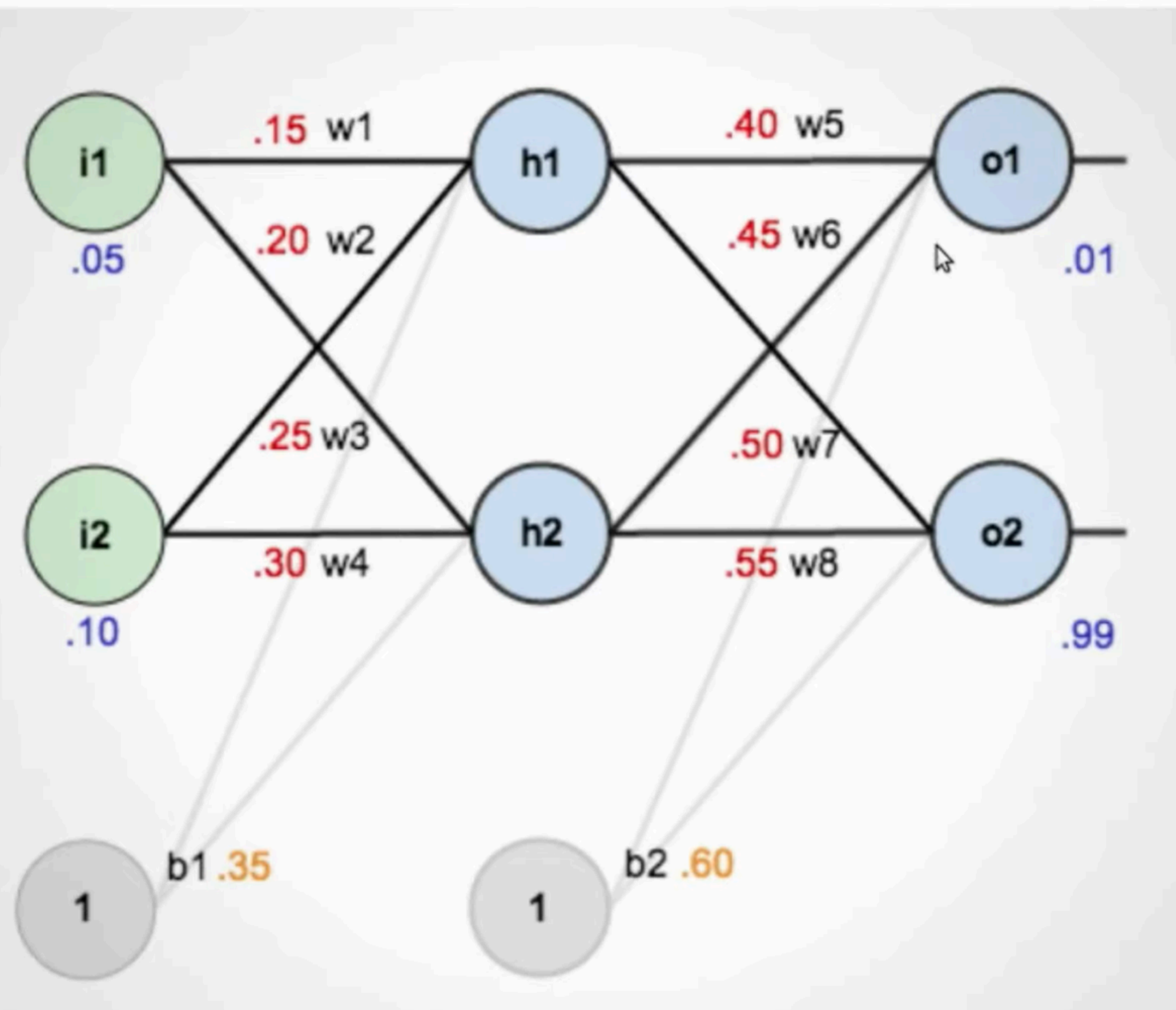
1

SIGMOID

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



- **RANGE:** [0, 1].
- **OUTPUT:** Good for Binary Classification (Cat vs. Not Cat).
- **HIDDEN LAYER DRAWBACK:** Severe Vanishing Gradient (<0.25)



$$i1=0.05, i2=0.10$$

$$w1=0.15, w2=0.20$$

$$w3=0.25, w4=0.30$$

$$b1=0.3$$

$$w5=0.4, w6=0.45$$

$$w7=0.5, w8=0.55$$

$$b2=0.6$$

$$o1=0.01, o2=0.99$$

We can now update w_1 :

$$w_1^+ = w_1 - \eta * \frac{\partial E_{total}}{\partial w_1} = 0.15 - 0.5 * 0.000438568 = 0.149780716$$

Vanishing Gradient Problem

Repeating this for w_2 , w_3 , and w_4

$$w_2^+ = 0.19956143$$

$$w_3^+ = 0.24975114$$

$$w_4^+ = 0.29950229$$

Why Deep Networks Fail? → The 2 Fundamental Problems

1. Vanishing Gradient Problem

👉 What happens?

During backpropagation, gradients (updates) become **very small** (close to 0).

📌 Why it happens:

- In deep neural networks, gradients are multiplied layer by layer
- With activation functions like **Sigmoid** or **Tanh**, derivatives are **very small** (≤ 1)
- Repeated multiplication → values shrink exponentially

🌟 Result:

- Early layers (closer to input) get **almost zero updates**
- Those layers **stop learning**

🎯 Real impact:

- Model learns only from last layers
- Deep networks become ineffective

Learning **STOP**

2. Exploding Gradient Problem

👉 What happens?

Gradients become **very large** (→ **infinity**) during backpropagation.

📌 Why it happens:

- When gradients > 1 are multiplied repeatedly
- Happens in deep networks or poor weight initialization

🌟 Result:

- Weights update becomes huge
- Model becomes unstable
- Loss may oscillate or become NaN

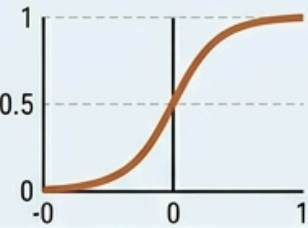


Learning becomes unstable

7 COMMONLY Used ACTIVATION Functions

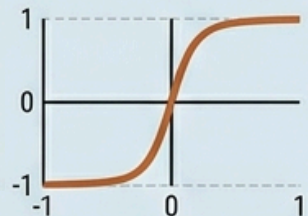
(Non-Linear)

1 SIGMOID

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$


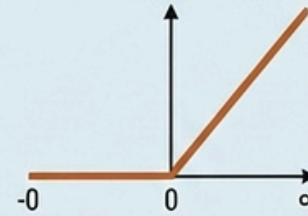
- **RANGE:** [0, 1].
- **OUTPUT:** Good for Binary Classification (Cat vs. Not Cat).
- **HIDDEN LAYER DRAWBACK:** Severe Vanishing Gradient <0.25)

2 TANH

$$f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$


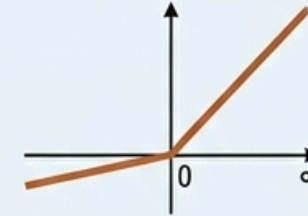
- **RANGE:** [-1, 1].
- **OUTPUT:** Zero-centered flow Classification (Cat vs. Not Cat).
- **HIDDEN LAYER DRAWBACK:** (Derivative max 0.25).

3 ReLU (Standard)

$$f(z) = \max(0, z)$$


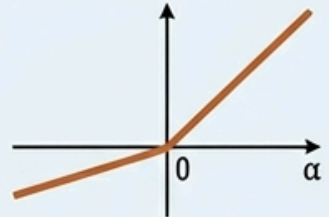
- **RANGE:** [0, ∞].
- **SOLVES:** Vanishing Gradient (Derivative 10 for positive input).
- **DRAWBACK:** "Dying ReLU" (Neurons with out, stop learning).

4 LEAKY ReLU

$$f(z) = \max(0.01z, z)$$


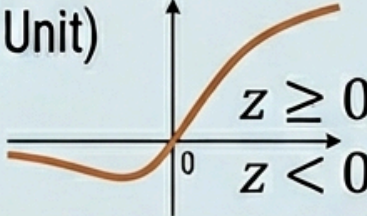
- **RANGE:** (-∞, ∞).
- **SOLVES:** "Dying ReLU" problem.
- **HOW:** Slight slope (e.g., 0.01).
- **HOW:** Slight slope (e.g., 0.01) for negative inputs.

5 PReLU (Parametric ReLU)

$$f(z) = \max(\alpha z, z)$$


- **LEARNABLE ALPHA:** Network learns the optimal negative slope.

6 ELU (Exponential Linear Unit)

$$f(z) = \begin{cases} z & z \geq 0 \\ \alpha(e^z - 1) & z < 0 \end{cases}$$


- **RANGE:** [-∞, ∞].
- **SOLVES:** Sharp ReLU corner.
- **ADVANTAGE:** Smooth near zero, controlled negative output.

7 SOFTMAX

USE: OUTPUT LAYER ONLY for Multi-Class Classification.

- **USE:** OUTPUT LAYER ONLY for Multi-Class Classification.
- **FUNCTION:** Converts scores into probability distribution.

Cat=5.1	→ Cat=0.92
Dog=1.2	→ Dog=0.03
Cat=3.3	→ Cat=0.03
Cat=2.3	→ Dog=0.01
Dog=1.3	→ Dog=0.03
Ct=....	Sums to 1.

Are you clear with

1. Why Deep Learning?
2. Inspiration behind Deep Neural Networks
3. ANN & Building Blocks of MLP?
4. Optimizers Need
5. Activation Functions
6. Loss Functions

Your Next Question should be...

How different Optimizers Work?

How Loss Functions Work?